# Joint Target Localization and Channel Estimation for ODDM-ISAC Systems

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***Abstract*—In pursuit of reliable performance for high-mobility integrated sensing and communication (ISAC) scenarios, the orthogonal delay-Doppler division multiplexing (ODDM) modulation can be leveraged to exploit the inherent channel sparsity in the delay-Doppler (DD) domain. In this letter, we propose a joint target localization and channel estimation method for ODDM-ISAC systems that employs a multi-pilot training frame to enhance parameter estimation with accumulated signal energy. By exploiting the block-circulant-like structure of the equivalent sampled DD domain (ESDD) channel matrix, multiple pilots are strategically arranged within a training frame. To facilitate effective energy accumulation from different pilots, a phase compensation strategy is devised, which improves the accuracy of parameter estimation for target localization and channel reconstruction. Moreover, the feasibility of the proposed method is theoretically analyzed when the actual delay and Doppler shift are on or off the quantized DD grid, respectively. Simulation results validate the effectiveness of the proposed method for both target localization and channel estimation.**

***Index Terms*—Channel estimation, integrated sensing and communication, orthogonal delay-Doppler division multiplexing, target localization.**

## I. Introduction

JOINT target localization and channel estimation has become an essential technology in integrated sensing and communication (ISAC) systems [1], [2], [3]. In particular, channel estimation performance can be improved by leveraging the location parameters of the sensing targets as supplementary spatial information [4], [5], [6], [7]. As one of the representative modulation techniques in ISAC, orthogonal frequency-division multiplexing (OFDM) provides reliable target localization and channel estimation in low-velocity scenarios [8], [9]. However, in high-mobility scenarios such as the Internet of vehicles and unmanned aerial vehicle (UAV) swarms, Doppler spread caused by moving targets disrupts subcarrier orthogonality [10], [11], making it challenging to accurately capture the channel response in OFDM-based ISAC systems.

To effectively characterize time-varying channels, it is advantageous to represent them in the delay-Doppler (DD) domain, such as those in the orthogonal time frequency space (OTFS) modulation [12], [13]. The parameters characterizing the DD domain channel remain approximately constant over periods much longer than the channel coherence time, enabling OTFS-based ISAC systems to operate robustly in high-mobility scenarios [14], [15]. Thus, efforts have been devoted to designing joint target localization and channel estimation methods for the OTFS-based ISAC systems [16], [17]. However, the core of OTFS modulation, namely the ideal pulse with bi-orthogonal robustness against time and frequency dispersions [12], is still unknown.

Recently, a DD domain multi-carrier (DDMC) modulation scheme named orthogonal delay-Doppler division multiplexing (ODDM) [18], [19] was designed based on the newly discovered DD domain orthogonal pulse (DDOP) [20]. Specifically, the DDOP comprises a train of square-root Nyquist pulses and maintains sufficient bi-orthogonality, enabling the modulation resolutions to match those of the channel in the DD domain [21]. This reduces the interference in the DD-domain received symbols [22], [23], making ODDM modulation well-suited for high-mobility scenarios and driving research into joint channel estimation and target localization for ODDM-ISAC systems. However, existing channel estimation methods mainly rely on a single high-power pilot in the DD domain [23], [24], [25], resulting in a high peak-to-average power ratio (PAPR) that challenges hardware implementation due to strict dynamic range requirements of power amplifiers [26]. Moreover, previous studies have focused on single-antenna receivers with limited localization capabilities.

In this letter, we propose a joint target localization and channel estimation method for multi-antenna ODDM-ISAC systems employing a multi-pilot training frame. Specifically, by exploiting the block-circulant-like structure of the equivalent sampled DD domain (ESDD) channel matrix in ODDM modulation, multiple pilots are arranged within a training frame to reduce the energy per pilot. Then, a phase compensation strategy is developed to tackle the phase deviation associated with received signals of pilots transmitted at different time instants and thus effectively accumulate the signal power. Additionally, the impact of the quantization mismatch between discrete DD grid

Received 20 December 2024; revised 6 February 2025; accepted 14 February 2025. Date of publication 18 March 2025; date of current version 16 April 2025. The work of Luning Lin, Hang Zheng, and Zhiguo Shi was supported in part by the National Natural Science Foundation of China under Grant U21A20456 and Grant 62271444 and in part by the Zhejiang Provincial Natural Science Foundation of China under Grant LZ23F010007. The work of Hai Lin was supported by JSPS KAKENHI under Grant 22H01491. The associate editor coordinating the review of this article and approving it for publication was Dr. Shunqiao Sun. *(Corresponding author: Zhiguo Shi.)*

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Digital Object Identifier 10.1109/LSP.2025.3552518

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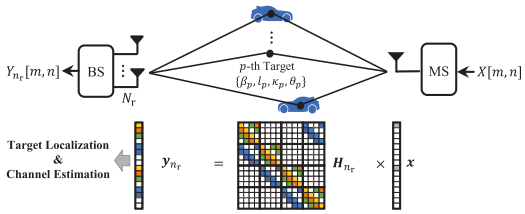


Fig. 1. Illustration of the deployed ISAC scenario.

points and actual DD parameters on channel reconstruction is analyzed, and the condition when such impact can be neglected is specified. The effectiveness of the proposed method is validated through numerical simulations.

## II. ODDM Signal Model

As depicted in Fig. 1, we consider an ODDM-ISAC system consisting of a base station (BS) and a mobile station (MS). The BS is equipped with a uniform linear array (ULA) of $N_{\mathrm{r}}$ antennas with the inter-element spacing being half of the signal wavelength $\lambda = c/f_c$, where $c$ denotes the light speed and $f_c$ denotes the carrier frequency. The MS is equipped with a single antenna. At the beginning of uplink transmission process, the MS transmits a training frame, which is reflected by multiple targets and received at the BS for joint target localization and channel estimation.

In ODDM modulation, each frame carries $MN$ information-bearing digital symbols $\{X[m,n]\}_{0\le m\le M-1, 0\le n\le N-1}$, where $[m,n]$ indicates the $m$-th multi-carrier (MC) symbol and the $n$-th subcarrier, $M$ and $N$ denote the numbers of MC symbols and subcarriers, respectively. The digital symbols are assigned according to the time-frequency or equivalently DD grid $\Gamma = \{m\frac{T}{M}, \dot{n}\frac{1}{NT}\}$ for $m = 0, 1, \ldots, M-1$, $\dot{n} = n - \frac{N}{2} = -\frac{N}{2}, \ldots, \frac{N}{2} - 1$, where $\frac{T}{M}$ and $\frac{1}{NT}$ are respectively the resolutions of time (delay) and frequency (Doppler), and $T$ is the interval between two adjacent pulses in the DDOP [21]. The transmitted ODDM waveform can be represented as

$$x(t) = \sum_{m=0}^{M-1} \sum_{\dot{n}=-N/2}^{N/2-1} X[m,[\dot{n}]_N]\gamma\left(t - m\frac{T}{M}\right) e^{\jmath 2\pi \frac{\dot{n}}{NT}(t - m\frac{T}{M})}, \tag{1}$$

where $\gamma(t) = \sum_{i=0}^{N-1} a(t - iT)$ denotes the DDOP with sufficient bi-orthogonality in the DD domain [20], $a(t)$ is a square-root Nyquist pulse with its Nyquist interval and time duration being respectively $\frac{T}{M}$ and $2Q\frac{T}{M}$, $Q \in \mathbb{Z}^+$ is the quality factor, and $[\cdot]_N$ is the modulo $N$ operator. The ODDM frame is extended by cyclic prefix (CP) and modulated onto the carrier wave to be transmitted through wireless channel.Considering that the transmitted signals are reflected by $P$ targets, the equivalent baseband signals received at the $n_{\mathrm{r}}$-th ($n_{\mathrm{r}} = 1, 2, \ldots, N_{\mathrm{r}}$) antenna of the BS is represented as[1]

$$y_{n_{\mathrm{r}}}(t) = \sum_{p=1}^{P} \beta_p x(t - \tau_p) e^{\jmath 2\pi v_p (t-\tau_p)} e^{-\jmath\pi(n_{\mathrm{r}}-1)\sin\theta_p} + z_{n_{\mathrm{r}}}(t), \tag{2}$$

where $\beta_p$, $\tau_p$, $v_p$, and $\theta_p$ denote respectively the gain, delay, Doppler shift, and direction-of-arrival (DOA) of the channel path corresponding to the $p$-th target, and $z_{n_{\mathrm{r}}}(t)$ is the baseband noise. Then, the signal component corresponding to the $[\dot{n}]_N$-th subcarrier of the $m$-th MC symbol can be obtained by receive filtering based on matched filters or correlators as

$$Y_{n_{\mathrm{r}}}[m,[\dot{n}]_N] = \int_{-\infty}^{+\infty} y_{n_{\mathrm{r}}}(t)\gamma^*\left(t - m\frac{T}{M}\right) e^{-\jmath 2\pi \frac{\dot{n}}{NT}(t - m\frac{T}{M})} dt, \tag{3}$$

where $(\cdot)^*$ denotes the conjugation operator. In practice, a low-complexity implementation of ODDM can be approximately achieved by $N$-point inverse discrete Fourier transform (IDFT) and discrete Fourier transform (DFT) when $2Q \ll M$ [23].

As such, the input-output relation of ODDM modulation can be represented as

$$\boldsymbol{y}_{n_{\mathrm{r}}} = \boldsymbol{H}_{n_{\mathrm{r}}}\boldsymbol{x} + \boldsymbol{z}_{n_{\mathrm{r}}}, \tag{4}$$

where $\boldsymbol{y}_{n_{\mathrm{r}}} = [\boldsymbol{y}_{n_{\mathrm{r}},0}^{\mathrm{T}}, \boldsymbol{y}_{n_{\mathrm{r}},1}^{\mathrm{T}}, \ldots, \boldsymbol{y}_{n_{\mathrm{r}},M-1}^{\mathrm{T}}]^{\mathrm{T}}$ is the received signal vector with $\boldsymbol{y}_{n_{\mathrm{r}},m} = [Y_{n_{\mathrm{r}}}[m,0], Y_{n_{\mathrm{r}}}[m,1], \ldots, Y_{n_{\mathrm{r}}}[m,N-1]]^{\mathrm{T}}$, $\boldsymbol{x} = [\boldsymbol{x}_0^{\mathrm{T}}, \boldsymbol{x}_1^{\mathrm{T}}, \ldots, \boldsymbol{x}_{M-1}^{\mathrm{T}}]^{\mathrm{T}}$ is the transmitted symbol vector with $\boldsymbol{x}_m = [X[m,0], X[m,1], \ldots, X[m,N-1]]^{\mathrm{T}}$, $\boldsymbol{H}_{n_{\mathrm{r}}}$ denotes the ESDD channel matrix of the $n_{\mathrm{r}}$-th receive antenna, $\boldsymbol{z}_{n_{\mathrm{r}}}$ is the additive white Gaussian noise vector, and $(\cdot)^{\mathrm{T}}$ denotes the transpose operator. The $(m,\tilde{m})$-th submatrix of $\boldsymbol{H}_{n_{\mathrm{r}}}$ is detailed in (5) shown at the bottom of this page, $\tilde{m} = 0, 1, \ldots, M-1$. In (5), $l_p \triangleq \frac{\tau_p}{T/M}$, $\kappa_p \triangleq v_p NT$ are defined respectively as the normalized delay and Doppler shift, $D$ denotes the normalized delay spread, $g(\tau) = \gamma(\tau) * \gamma^*(-\tau)$ is the effective pulse characterizing the overall effects of the filters at the MS and the BS,[2] $\boldsymbol{C}_p$ denotes the Doppler spread matrix with its $(n,\tilde{n})$-th element being

$$[\boldsymbol{C}_p]_{(n,\tilde{n})} = \frac{1}{N}\frac{1 - e^{\jmath 2\pi(\tilde{n}-n+\kappa_p)}}{1 - e^{\jmath\frac{2\pi}{N}(\tilde{n}-n+\kappa_p)}}, \quad n, \tilde{n} = 0, 1, \ldots, N-1, \tag{6}$$

and $\boldsymbol{\Psi} = \mathrm{diag}\{1, e^{-\jmath 2\pi/N}, \ldots, e^{-\jmath 2\pi(N-1)/N}\}$ is a diagonal matrix with extra phase shift for CP symbols on its diagonal. In particular, the circulant property within each submatrix in $\boldsymbol{H}_{n_{\mathrm{r}}}$ originates from the periodicity of the Dirichlet defining $\boldsymbol{C}_p$ in (5), which characterizes the ICI within each MC symbol. Meanwhile, the block-circulant-like structure across submatrices is associated with $g(\tau)$, which describes the interference among MC symbols, the CP, and the phase shifts varying with the MC symbols [23].

[1] We consider the case where the line-of-sight (LoS) path is absent, while the extension to the case with a LoS path is straightforward.

[2] Here, we assume that the impact of channel variations on $g(\tau)$ is negligible, which is reasonable in the case of low Doppler spreads [23].

$$\boldsymbol{H}_{n_{\mathrm{r}}}^{(m,\tilde{m})} = \begin{cases} \sum_{p=1}^{P} \beta_p g\left((m - \tilde{m} - l_p)\frac{T}{M}\right) e^{\jmath 2\pi \frac{(m-l_p)\kappa_p}{MN}} e^{-\jmath\pi(n_{\mathrm{r}}-1)\sin\theta_p} \boldsymbol{C}_p, & \text{if } 0 \le m - \tilde{m} \le D-1, m \ge D-1, \text{or } 0 \le \tilde{m} \le m, m < D-1, \\ \sum_{p=1}^{P} \beta_p g\left((m - \tilde{m} - l_p)\frac{T}{M}\right) e^{\jmath 2\pi \frac{(m-l_p)\kappa_p}{MN}} e^{-\jmath\pi(n_{\mathrm{r}}-1)\sin\theta_p} \boldsymbol{C}_p\boldsymbol{\Psi}, & \text{if } M + m - D + 1 \le \tilde{m} \le M-1, m < D-1, \\ \boldsymbol{0}, & \text{otherwise.} \end{cases} \tag{5}$$

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## III. Joint Target Localization and Channel Estimation

### A. Multi-Pilot Frame Structure

According to the block-circulant-like structure of the ESDD channel matrix $\boldsymbol{H}_{n_\mathrm{r}}$ in (5), each row and column in $\boldsymbol{H}_{n_\mathrm{r}}$ is associated with the delay and Doppler shift of $P$ targets. Since the delay, Doppler, and angle parameters of multiple channel paths remain approximately invariant within each DD domain frame [27], a multi-pilot training frame can be designed to enhance parameter estimation by probing the DD domain channel at different time instants, in contrast to relying on a single high-power pilot. Specifically, $U$ pilots are transmitted at an interval of $L$ in an ODDM frame as

$$X[m,n]=\begin{cases}\alpha, & m=uL, n=0,\\ 0, & \text{otherwise}\end{cases} \tag{7}$$

for $u=0,1,\ldots,U-1$, and $U\leq\lfloor\frac{M}{L}\rfloor-1$. Here, $\lfloor\cdot\rfloor$ denotes the rounding down operator. For notational simplicity, we assume $\alpha=1$ in the following. Note that, the interval $L$ in (7) is set larger than the delay spread $D$ in (5) to prevent the received signals of different pilots from being overlapped.

As such, the received signal vector associated with the training frame in (7) is composed of $U$ non-overlapping signal subvectors. Specifically, the signal subvector corresponding to the $u$-th pilot is denoted as $\tilde{\boldsymbol{y}}_{n_\mathrm{r},u}\triangleq[\boldsymbol{y}_{n_\mathrm{r}}]_{(uNL:(u+1)NL-1)}$, where $[\cdot]_{(i_1:i_2)}$ represents the slice of a vector comprising elements indexed from $i_1$ to $i_2$. By substituting (7) and (5) into (4), the $q$-th element in $\tilde{\boldsymbol{y}}_{n_\mathrm{r},u}$ can be derived as

$$\begin{aligned}\left[\tilde{\boldsymbol{y}}_{n_\mathrm{r},u}\right]_{(q)}&=\sum_{p=1}^{P}\beta_p g\left(\left(\left\lfloor\frac{q}{N}\right\rfloor-l_p\right)\frac{T}{M}\right)e^{\jmath 2\pi\frac{\left(uL+\left\lfloor\frac{q}{N}\right\rfloor-l_p\right)\kappa_p}{MN}}\\&\times e^{-\jmath\pi(n_\mathrm{r}-1)\sin\theta_p}\frac{1}{N}\frac{1-e^{\jmath 2\pi(\kappa_p-[q]_N)}}{1-e^{\jmath\frac{2\pi}{N}(\kappa_p-[q]_N)}}+\left[\boldsymbol{z}_{n_\mathrm{r}}\right]_{(uNL+q)}\end{aligned} \tag{8}$$

for $q=0,1,\ldots,NL-1$, where $[\cdot]_{(q)}$ denotes the $q$-th element in a vector, and $\frac{1}{N}\frac{1-e^{\jmath 2\pi(\kappa_p-[q]_N)}}{1-e^{\jmath\frac{2\pi}{N}(\kappa_p-[q]_N)}}$ is the Dirichlet kernel characterizing the leakage in the Doppler domain. In (8), since the effective pulse $g(\tau)$ is a Nyquist pulse, $g((\lfloor\frac{q}{N}\rfloor-l_p)\frac{T}{M})$ reaches its maximum when $\lfloor\frac{q}{N}\rfloor$ approaches $l_p$ and exhibits relatively small values otherwise. Meanwhile, the Dirichlet kernel has the maximum value when $\kappa_p$ approaches $[q]_N$, which is demonstrated by

$$\begin{aligned}&\lim_{\kappa_p-[q]_N\to 0}\frac{1}{N}\frac{1-e^{\jmath 2\pi(\kappa_p-[q]_N)}}{1-e^{\jmath\frac{2\pi}{N}(\kappa_p-[q]_N)}}\\&=\lim_{\kappa_p-[q]_N\to 0}\frac{1}{N}\frac{\jmath e^{\jmath 2\pi(\kappa_p-[q]_N)}}{\frac{1}{N}\jmath e^{\jmath\frac{2\pi}{N}(\kappa_p-[q]_N)}}=1.\end{aligned} \tag{9}$$

When $\kappa_p$ deviates from $[q]_N$, the value of the Dirichlet kernal can be much smaller.

### B. Phase Compensation for Delay and Doppler Estimation

Based on the above properties of the effective pulse and Dirichlet kernel, the delay and Doppler of sensing targets are associated with the magnitudes of elements in each signal subvector $\tilde{\boldsymbol{y}}_{n_\mathrm{r},u}$. Thus, summing these subvectors may enhance the signal strength for delay and Doppler estimation.

However, since the pulses associated with different pilots are transmitted at various time instants, there is a phase deviation term $e^{\jmath 2\pi\frac{uL\kappa_p}{MN}}$ in the received signal subvector $\tilde{\boldsymbol{y}}_{n_\mathrm{r},u}$, resulting in potential signal cancellation when summing multiple subvectors. To address this, a phase compensation matrix

$$\boldsymbol{W}_u=\mathrm{diag}\left(\left[e^{\jmath 2\pi\frac{uL\cdot[0]_N}{MN}},e^{\jmath 2\pi\frac{uL\cdot[1]_N}{MN}},\ldots,e^{\jmath 2\pi\frac{uL\cdot[NL-1]_N}{MN}}\right]\right) \tag{10}$$

is designed to mitigate the phase deviation in the $u$-th subvector, where $\mathrm{diag}(\cdot)$ is the diagonal operator.

Then, a magnitude function is defined to facilitate extracting delay and Doppler parameters from the received signals as

$$\Phi_{n_\mathrm{r}}(q)=\frac{1}{U}\sum_{u=0}^{U-1}[\boldsymbol{W}_u^\mathrm{H}\tilde{\boldsymbol{y}}_{n_\mathrm{r},u}]_{(q)}, \tag{11}$$

where $(\cdot)^\mathrm{H}$ denotes the Hermitian transpose operator.

*Theorem 1:* In the case when delay and Doppler shift lie on the DD grids in $\Gamma$, i.e., $l_p,\kappa_p\in\mathbb{Z}$, the phase compensation matrix $\boldsymbol{W}_u$ defined in (10) maximizes the signal-to-noise ratio (SNR) in the magnitude function $\Phi_{n_\mathrm{r}}(q)$ in (11) at the index $q=q_p\triangleq Nl_p+\kappa_p\in\{0,1,\ldots,NL-1\}$.

*Proof:* In the on-grid case, $\Phi_{n_\mathrm{r}}(q)$ can be reformulated by substituting (8) into (11) as

$$\Phi_{n_\mathrm{r}}(q)=\begin{cases}\frac{1}{U}\sum_{u=0}^{U-1}\left[\boldsymbol{W}_u^\mathrm{H}\boldsymbol{s}_{n_\mathrm{r},u}\right]_{(q)}+[\tilde{\boldsymbol{z}}_{n_\mathrm{r},u}]_{(q)}, & \text{if } q=q_p\in\{q_1,\ldots,q_P\},\\ [\tilde{\boldsymbol{z}}_{n_\mathrm{r},u}]_{(q)}, & \text{otherwise},\end{cases} \tag{12}$$

where $[\boldsymbol{W}_u^\mathrm{H}\boldsymbol{s}_{n_\mathrm{r},u}]_{(q)}=\beta_p g(0)e^{-\jmath\pi(n_\mathrm{r}-1)\sin\theta_p}e^{\jmath 2\pi\frac{uL(\kappa_p-[q]_N)}{MN}}$, and $\tilde{\boldsymbol{z}}_{n_\mathrm{r},u}=\frac{1}{U}\sum_{u=0}^{U-1}\boldsymbol{W}_u^\mathrm{H}\boldsymbol{z}_{n_\mathrm{r},u}$ denotes the equivalent noise vector. According to the Cauchy–Schwarz inequality [28], the SNR $\frac{|\sum_{u=0}^{U-1}[\boldsymbol{W}_u^\mathrm{H}\boldsymbol{s}_{n_\mathrm{r},u}]_{(q)}|^2}{\mathbb{E}\{|[\tilde{\boldsymbol{z}}_{n_\mathrm{r},u}]_{(q)}|^2\}}$ of $\Phi_{n_\mathrm{r}}(q)$ is maximized at the index $q=q_p$ when the exponential term $e^{\jmath 2\pi\frac{uL(\kappa_p-[q_p]_N)}{MN}}$ in the signal components across different $u$ equals 1. Here, $\mathbb{E}\{\cdot\}$ denotes the expectation operator. Since $[q_p]_N=[Nl_p+\kappa_p]_N=\kappa_p$, we have $e^{\jmath 2\pi\frac{uL(\kappa_p-[q_p]_N)}{MN}}=1$, leading to maximum SNR for $\Phi_{n_\mathrm{r}}(q_p)$. ■

On that basis, by employing the constant false alarm rate (CFAR) technique for peak detection in $\Phi_{n_\mathrm{r}}$ [29], [30], the delay and Doppler shift of the $p$-th target can be estimated as

$$\hat{\tau}_p=\frac{T}{M}\left\lfloor\frac{q_p}{N}\right\rfloor,\quad \hat{\upsilon}_p=\frac{1}{NT}[q_p]_N. \tag{13}$$

In the following, joint target localization and channel estimation is conducted in the on-grid case, while the impact of off-grid delay and Doppler will be discussed in Section IV.

### C. On-Grid Target Localization and Channel Reconstruction

To determine the coordinate of the $p$-th target, its DOA can be estimated by exploiting the phase difference of the signals from various antennas. In particular, a signal vector containing the signals received from all antennas can be constructed as

$$\dot{\boldsymbol{y}}_{q_p,u}=\left[[\tilde{\boldsymbol{y}}_{1,u}]_{(q_p)},[\tilde{\boldsymbol{y}}_{2,u}]_{(q_p)},\ldots,[\tilde{\boldsymbol{y}}_{N_\mathrm{r},u}]_{(q_p)}\right]^\mathrm{T}. \tag{14}$$

By substituting (8) into (14), we have

$$\dot{\boldsymbol{y}}_{q_p,u}=\beta_p g(0)e^{\jmath 2\pi\frac{uL\kappa_p}{MN}}\boldsymbol{a}(\theta_p)+\dot{\boldsymbol{z}}_{q_p,u}, \tag{15}$$

where $\boldsymbol{a}(\theta_p)=[1,e^{-\jmath\pi\sin\theta_p},\ldots,e^{-\jmath\pi(N_\mathrm{r}-1)\sin\theta_p}]^\mathrm{T}$ denotes the steering vector associated with the $p$-th target, and $\dot{\boldsymbol{z}}_{q_p,u}$ is the corresponding noise vector. Then, the DOA of the $p$-th target $\theta_p$ can be obtained through gridless parameter estimation

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methods like ESPRIT [31]. Different from the on-grid estimates of delay and Doppler shift, gridless DOA estimation is obtained to enhance the accuracy of target localization.

Based on the estimates $\hat{\tau}_p$ and $\hat{\theta}_p$, the location of the $p$-th target relative to the BS can be calculated as $[\hat{x}_p, \hat{y}_p]^{\mathrm{T}} = \frac{c\hat{\tau}_p \sin(\hat{\phi}_p)}{\sin(\hat{\theta}_p)+\sin(\hat{\phi}_p)}[\sin(\hat{\theta}_p), \cos(\hat{\theta}_p)]^{\mathrm{T}}$. Here, $\hat{\phi}_p$ denotes the angle-of-departure estimated according to the geometrical structure among the MS, the BS, and the $p$-th target [32]. Finally, to reconstruct the ESDD channel matrix, the complex gain $\hat{\beta}_p$ of the $p$-th path can be estimated as

$$\hat{\beta}_p = \sum_{u=0}^{U-1} \frac{e^{-\jmath 2\pi \frac{uL\hat{\kappa}_p}{MN}} \boldsymbol{a}^{\mathrm{H}}(\hat{\theta}_p)\dot{\boldsymbol{y}}_{q_p,u}}{UN_{\mathrm{r}}g(0)}. \tag{16}$$

The ESDD channel matrix $\hat{\boldsymbol{H}}_{n_{\mathrm{r}}}$ can then be reconstructed by substituting the estimated parameters $\{\hat{\tau}_p, \hat{v}_p, \hat{\theta}_p, \hat{\beta}_p\}$ into (5). Since the received signal subvectors of different transmitted pilots are effectively utilized through the proposed phase compensation strategy, enhanced accuracy for joint target localization and channel estimation can be achieved.

## IV. Impact of Off-Grid Delay and Doppler Shift

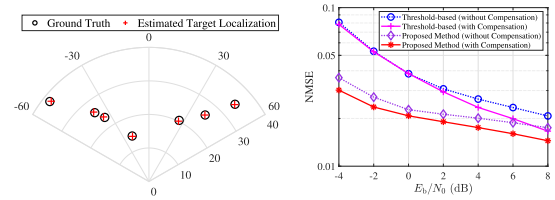
In this section, we analyze the feasibility of the proposed method in the case when delay and Doppler shift are off the DD grids. In this case, quantization errors in delay and Doppler shift estimation are associated with signal bandwidth and the duration of a frame, respectively. Off-grid delay and Doppler shift contribute to the channel spreading along the delay and Doppler domains,[3] resulting in $\breve{P}$ ($\breve{P} > P$) equivalent on-grid paths characterized by $\{\breve{l}_{\breve{p}}, \breve{\kappa}_{\breve{p}}, \breve{\theta}_{\breve{p}}, \breve{\beta}_{\breve{p}}\}$, $\breve{p} = 1, 2, \ldots, \breve{P}$. By estimating the parameters of each equivalent on-grid path, the corresponding channel matrix $\hat{\boldsymbol{H}}_{n_{\mathrm{r}}}$ can be reconstructed.

*Theorem 2:* When $N$ is sufficiently large, the reconstructed channel matrix $\hat{\boldsymbol{H}}_{n_{\mathrm{r}}}$ based on the dominant equivalent on-grid parameters $\{\breve{l}_{\breve{p}}, \breve{\kappa}_{\breve{p}}\}$ approaches the actual channel matrix $\boldsymbol{H}_{n_{\mathrm{r}}}$ with the parameters $\{l_p, \kappa_p\}$.

*Proof:* In the case of $P = 1$, we have $[\boldsymbol{H}_{n_{\mathrm{r}}}^{(m,\tilde{m})}]_{(n,\tilde{n})} = e^{\jmath 2\pi \frac{\kappa_1}{MN}}[\boldsymbol{H}_{n_{\mathrm{r}}}^{([m-1]_M,[\tilde{m}-1]_M)}]_{(n,\tilde{n})}$ according to (5). Meanwhile, for the reconstructed channel matrix, we have $[\hat{\boldsymbol{H}}_{n_{\mathrm{r}}}^{(m,\tilde{m})}]_{(n,\tilde{n})} = e^{\jmath 2\pi \frac{\breve{\kappa}_{\breve{p}}}{MN}}[\hat{\boldsymbol{H}}_{n_{\mathrm{r}}}^{(m-1,\tilde{m}-1)}]_{(n,\tilde{n})}$ if $\exists \breve{\kappa}_{\breve{p}} = [n-\tilde{n}]_N$. Then, the largest value of the ratio between $[\boldsymbol{H}_{n_{\mathrm{r}}}^{(m,\tilde{m})}]_{(n,\tilde{n})}$ and $[\hat{\boldsymbol{H}}_{n_{\mathrm{r}}}^{(m,\tilde{m})}]_{(n,\tilde{n})}$ can be represented by $e^{\jmath \frac{2\pi}{N}\max\{|\breve{\kappa}_{\breve{p}}-\kappa_1| | \breve{\kappa}_{\breve{p}} \in \mathcal{K}\}}$, where $\max\{\cdot\}$ denotes the maximum operator, and $\mathcal{K} = \{\breve{\kappa}_1, \breve{\kappa}_2, \ldots, \breve{\kappa}_{\breve{P}}\}$. Based on the characteristics of Dirichlet kernel discussed in Section III-A, the majority of energy in received signals is retained by a few subcarriers with indices near $\kappa_1$ [23]. As such, $\frac{\max\{|\breve{\kappa}_{\breve{p}}-\kappa_1| | \breve{\kappa}_{\breve{p}} \in \mathcal{K}\}}{N}$ is approximately 0 considering large $N$, leading to insignificant differences between the actual and reconstructed channel matrices. The above analysis can be extended to the case of $P > 1$, since the phase rotation coefficient $e^{\jmath 2\pi \frac{\kappa_p}{MN}}$ of each off-grid path would approximate those of the corresponding on-grid paths when $N$ is sufficiently large.

Given that the value of $N$ can be large in practical systems [33], the estimation of on-grid parameters can substitute the actual values, supporting the feasibility of channel reconstruction via the proposed method in the off-grid cases.

[3] The channel spreading does not introduce ghost targets since only the local maximums at grid points closest to the true off-grid parameters are identified as peaks for target localization.




(a) (b)

Fig. 2. Illustration of the performance of target localization and channel estimation. (a) Target localization. (b) Channel estimation.

## V. Simulation Results

In this section, numerical results are presented to illustrate the performance of the proposed method in the ODDM-ISAC system. The quadrature phase-shift keying modulation (QPSK) is employed to generate the DD domain information symbols. The carrier frequency $f_c$ is set to 28 GHz, and the subcarrier spacing is $\frac{1}{NT} = 15$ kHz. The numbers of transmitted pilots, symbols per time frame, subcarriers, and antennas at the BS are set to $U = 9$, $M = 256$, $N = 32$, and $N_{\mathrm{r}} = 8$, respectively. A square-root-raised-cosine pulse is employed as $a(t)$, with the roll-off factor of $\rho = 0.1$ and quality factor of $Q = 10$. We consider $P = 7$ targets with the parameters $[\theta_1, \theta_2, \ldots, \theta_7] = [-20.2^\circ, 48.1^\circ, -38.2^\circ, -51.1^\circ, -34.6^\circ, 0.3^\circ, 26.3^\circ]$, $[c\tau_1, c\tau_2, \ldots, c\tau_7] = [56.3, 43.8, 49.2, 41.3, 51.3, 53.7, 46.2]$ m, and $[v_1, v_2, \ldots, v_7] = [0.45, 1.65, 1.35, 15.03, 1.2, 0.15, 13.8]$ kHz. The path gain $\beta_p$ is randomly selected from a unit circle. To validate the effectiveness of the target localization in the proposed method, the range-angle map viewed from the BS is demonstrated in Fig. 2(a), where the pilot-to-noise ratio (PNR) is set to $E_{\mathrm{b}}/N_0 = 4$ dB, and the probability of false alarm in the CFAR detection is set to $5 \times 10^{-3}$. It can be observed that each target can be effectively identified and localized.

In Fig. 2(b), the channel estimation accuracy of the proposed method is compared to that of the conventional threshold-based method [24] in the cases with and without the phase compensation strategy in Section III. The normalized mean squared error (NMSE) of $\boldsymbol{H}_{n_{\mathrm{r}}}$ is employed as the performance metric. The proposed method achieves superior accuracy in channel estimation compared to the threshold-based method, as the latter does not explicitly estimate the location parameters of the physical channel. In contrast, the proposed method estimates these parameters to both enable target location functionality and enhance channel estimation accuracy. Moreover, the phase compensation strategy further improves accuracy by leveraging the unique structure of the ESDD channel matrix to enhance the received signal strength for each channel path.

## VI. Conclusion

In this letter, we proposed a joint target localization and channel estimation method for ODDM-ISAC systems with multi-pilot training frames. By leveraging the block-circulant-like structure of the ESDD channel matrix, a phase compensation strategy is developed to achieve enhanced performance of target localization and channel reconstruction. Numerical results validate the effectiveness of the proposed method.

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